

Gabriel Faes

Technical Designer & Programmer

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Games Engineering MSc developer with a Mathematics & Computer Science background, experience on shipped commercial DLC, and strong foundations in C++, C#, and Python. Created projects in Unreal Engine, Unity, and made a game engine and game with C++ using DirectX12.

Skills: C++, C#, Python, SQL, Unity, Unreal, Source Control (Git), REST APIs, Computer Graphics (DirectX12), Rating System Design, Audio Engineering, Machine Learning, HTML/CSS.

Strong ability to communicate, fast learner, highly developed sense of logic, and very detail minded.

WORK EXPERIENCE

Additional Level Designer, Brace Yourself Games, Canada (Remote Freelance) Feb 2025 – Present

- Helped during development as a playtester, and then as Additional Level Designer for official DLC content.
- Designed official DLC rhythm charts by taking into account musical structure, beat and rhythmic patterns, readability, as well as the required chart difficulty.
- Currently 12 released charts on songs from collaborations with [Pizza Tower](#), [Spin Rhythm XD](#), and [Undertale](#).
- Worked with the team to ensure quality and playability, iterating the design through feedback, balancing challenge, fairness, flow, and playability, while adapting to remote team communication and contributing to on-time project delivery.

Game Night Organizer & Host, La Médiathèque de Monaco, Monaco (Freelance) 2024 – 2025

- Prepared, organized and then hosted a trivia night inspired by the French game show “Le Grand Concours”.
- Devised 400+ trivia questions, and structured them into the game’s format using PowerPoint, Python, and Unity, ensuring all player-facing interactions are as intuitive as possible.
- Hosted the game night, presenting rules and adapting pacing and tone for the different ages and players present.

SPECIFIC PROJECTS

Dinosaur Pottery Rhythm Game | C++, DirectX12

- Made a [short rhythm game](#) using only C++ and the DirectX12 API.
- Coded all the implementations for an animated model, custom HLSL shaders (with their buffers in DirectX12), spawning models and instances with the DirectX12 API, and GPU draw calls.
- Coded the gameplay, including player and camera movement, a short tutorial, dynamic voice lines, pot throwing schedule, feedback when hit by a pot, and overall results at the end of the game.

Relativistic Space Race | Unity, C#

- Created a [time-trial racing prototype](#) set in space implementing Einstein’s special theory of relativity.
- Every object stores its spacetime curve, from which they can be accurately distorted with time and length dilation using the Lorentz equations. The code was then further optimized to run smoothly in real-time.

UK Game Show Adaptation | Unity, C#

- Built a playable Unity adaptation of the UK game show “1000 Heartbeats”, including 15 different rounds, adaptive music and full gameplay logic.

Glicko Rating – SpyParty Competitive Match Results | Python

- Built Python tooling to process competitive match results and generate predicted player skill ratings using the Glicko rating model and used this code to analyze over 5000 SpyParty competitive matches (to date).
- Made the results [viewable in a spreadsheet](#) which has gone on to be used for various purposes, including as a match history tool, and as the main starting point for ranking players in virtually all competitive tournaments.

Tournament Design, Organization, and Casting

- Designed and ran multiple tournament formats across games, including Rift of the NecroDancer, SpyParty and Project Winter; formats included custom charts, anonymised brackets, group stages, double elimination and live casting logistics.
- Coordinated rules, player communication, score tracking, submissions, volunteer artists/designers, stream assets and casting schedules.

Game Localization – Keep Talking and Nobody Explodes

- Contributed to the French localization of Keep Talking and Nobody Explodes, translating both in-game content and the bomb-defusal manual while preserving context, rules clarity and puzzle logic, something particularly difficult due to how the game interacts with the players' verbal communication (such as the game using homonyms to confuse players).

EDUCATION

MSc in Games Engineering, University of Warwick, England

Sep 2025 – Sep 2026

- Created [a game in Unreal Engine](#) using C++ in a team of six in the Games Engine Design and Development module. Coordinated the team, designed the main game loop, added the voice lines, tuned the entire game (requiring communication with all other team members) to feel right in terms of difficulty and scale, and helped in many other smaller ways.
- Modules: Programming and Fundamental Algorithms, Advanced Computer Graphics, Games Engine Design and Development, and Fundamentals of Game Research, Development and Management.
- Dissertation in Rating Systems and literature review in [Special Relativity for Games](#).

BSc in Mathematics & Computer Science, École Polytechnique, France

Sep 2022 – July 2025

- 197 ECTS credits with an average of 13.9/20.
- Subjects: Computer Graphics, Concurrent Computing, Measure, Statistics, Functional Programming, Compilers, Complex Analysis, Networks, Formal Languages, Euclidian and Hermitian Spaces, and more.
- Bachelor thesis on [mathematical tournament design, comparing their fairness and efficacy](#).

International Baccalaureate Diploma, Hockerill Anglo-European College, England

Sep 2020 – July 2022

- Final Grade of 37/42 studying Maths, Computer Science, Physics, Economics, English, and Spanish.
- Won a Gold certificate in UKMT (Maths Olympics) and Seacole Student of the Year.

LEADERSHIP, CREATIVITY & COMMUNICATION

- **MUN:** Attended numerous MUN conferences, both as a delegate and later as a Student Officer, and became president of my MUN student group, organising debates, resolution writing sessions, and more.
- **Music:** Board Member of the Bachelor Music Committee (Polytechnique), coordinating Christmas and special-event concerts; classically trained pianist (conservatoire ages 5-11; 15+ years playing), with strong interest in video-game music, jazz and composition.
- **GameX:** President of the Game Committee (Polytechnique) organizing recurring game nights and tournaments, coordinating participants, formats, and event logistics.

LANGUAGES

- **Linguistic:** English (C2), French (C2), Spanish (C1), German (A2), Japanese (A1)